

(b) the subgraphs  $G_{i \leftarrow r}(t)$ .

Similarly, define the subgraph  $G_i(t)$  to be the subgraph using all  $r \in N_{\text{out}}(i)$ .

3) *Input update*: For all  $(i, j) \in E$ , let  $G_{i \leftarrow j}(t+1)$  be the subgraph containing the node  $x_j$  and for all  $\ell \in N_{\text{in}}(j) \neq i$ :

- (a) the edges  $(\ell, j)$ ;
- (b) the output nodes  $y_\ell$ ; and
- (c) the subgraphs  $G_{\ell \rightarrow j}(t)$ .

Similarly, define the subgraph  $G_j(t)$  to be the subgraph using all  $\ell \in N_{\text{in}}(j)$ . Set  $t = t+1$  and return to step 1.

Now let  $H_{i \rightarrow j}(t)$  be the sigma algebra generated by the components  $y_\ell$  contained in the computation subgraph  $G_{i \rightarrow j}(t)$ . Similarly, let  $H_{i \leftarrow j}(t)$  be the sigma algebra generated by the components  $y_\ell$  contained in the computation subgraph  $G_{i \leftarrow j}(t)$ . To analyze the BP algorithm with the “genie” initialization in Appendix B, let  $H_{i \rightarrow j}^{\text{genie}}(t)$  and  $H_{i \leftarrow j}^{\text{genie}}(t)$  be respectively the sigma algebras generated by the entire vector  $\mathbf{y}$  and the variable  $x_r$  not in the computation subgraphs  $G_{i \rightarrow j}(t)$  and  $G_{i \leftarrow j}(t)$ . The following results are standard for BP and can be proven using arguments as in [41].

*Lemma 4*: Consider the sigma algebras  $H_{i \leftarrow j}(t)$  and  $H_{i \rightarrow j}(t)$  defined above.

- (a) In the standard BP algorithm in Section III, the distributions  $p_{i \leftarrow j}^x(t, x_j)$  are  $H_{i \leftarrow j}(t)$  measurable, and the distribution  $p_{i \rightarrow j}^z(t, z_i)$  and likelihood function  $p_{i \rightarrow j}^u(t, u_j)$  are  $H_{i \rightarrow j}(t)$  measurable.
- (b) In the relaxed BP algorithm,  $\hat{x}_{i \leftarrow j}(t)$  and  $\hat{q}_{i \leftarrow j}(t)$  are  $H_{i \leftarrow j}(t)$  measurable and  $\hat{z}_{i \rightarrow j}(t)$  and  $\hat{u}_{i \rightarrow j}(t)$  are  $H_{i \rightarrow j}(t)$  measurable.

Similarly, under the oracle initialization described in Appendix B, the above statements hold with  $H_{i \leftarrow j}(t)$  and  $H_{i \rightarrow j}(t)$  replaced by  $H_{i \leftarrow j}^{\text{genie}}(t)$  and  $H_{i \rightarrow j}^{\text{genie}}(t)$ .

*Lemma 5*: Consider the standard BP algorithm in Section III and the computation subgraphs defined above.

- (a) If  $G_{i \leftarrow j}(t)$  is a tree, then  $p_{i \leftarrow j}^x(t, x_j)$  is the conditional distribution of  $x_j$  given  $H_{i \leftarrow j}(t)$ .
- (b) If  $G_{i \rightarrow j}(t)$  is a tree, then  $p_{i \rightarrow j}^z(t, z_i)$  is the conditional distribution of  $z_i$  given  $H_{i \rightarrow j}(t)$ .

Similarly, under the genie initialization, the above statements hold with  $H_{i \leftarrow j}(t)$  and  $H_{i \rightarrow j}(t)$  replaced by  $H_{i \leftarrow j}^{\text{genie}}(t)$  and  $H_{i \rightarrow j}^{\text{genie}}(t)$ .

*Lemma 6*: Consider the relaxed BP algorithm in Section IV under the assumptions in Section V.

- (a) If  $G_{i \leftarrow j}(t)$  is a tree, then all the terms  $(\hat{z}_{\ell \rightarrow j}(t), \mu_{i \rightarrow j}^z(t))$  are independent for different values  $\ell \in N_{\text{in}}(j) \neq i$ .
- (b) If  $G_{i \rightarrow j}(t)$  is a tree, then the random vectors  $\theta_{i \leftarrow r}^x(t)$  are independent for different values  $r \in N_{\text{out}}(i) \neq j$ .

## APPENDIX E PROOF OF THEOREM 1(A)

We prove this by induction. It is clear that (42a) holds for  $t = 1$ . In part B below we will show that if (42a) holds for some  $t$ , then so does (42c) and (42d). Then, in part C, we will show that if (42c) holds for some  $t$ , then (42a) holds for  $t+1$ . This will complete the induction argument. Part A provides a preliminary calculation that we need in part C.

### A. Derivatives of the Score Function

The following lemma characterizes the derivatives  $D_r(y, \hat{z}, \mu)$  in (16). The result can also be found in [14], but we sketch the proof here for completeness. We will use this result below in the analysis of the output node update.

*Lemma 7*: Fix  $\hat{z}$  and  $\mu$  and consider random variables  $y$  and  $z$  generated by  $z \sim \mathcal{N}(\hat{z} + u, \mu)$  and  $y \sim p_{Y|Z}(y|z)$  for some  $u \in \mathbb{R}$ . Consider the derivative of the score function in (16). Then,

$$\mathbf{E}[D_1(y, \hat{z}, \mu)|u] = -u\mathbf{E}[D_2(y, \hat{z}, \mu)|u=0] + O(u^2), \quad (51a)$$

$$\mathbf{var}[D_1(y, \hat{z}, \mu)|u] = \mathbf{E}[D_2(y, \hat{z}, \mu)|u=0] + O(u^2). \quad (51b)$$

*Proof*: To simplify the notation, we will drop the dependence on  $\hat{z}$  and  $\mu$ . So, we will write, for example,  $D_1(y)$  for  $D_1(y, \hat{z}, \mu)$ . To prove (51a), we first note that

$$\begin{aligned} \mathbf{E}[D_1(y)|u] &= \int p_{Y|U}(y|u) D_1(y) dy \\ &= \int p_{Y|U}(y|0) D_1(y) dy \\ &\quad + u \int \frac{\partial}{\partial u} p_{Y|U}(y|u) \Big|_{u=0} D_1(y) dy + O(u^2). \end{aligned} \quad (52)$$

Now, for the first term in (52), note that using the definition of  $D_1(y)$  in (16), we have

$$\begin{aligned} &\int p_{Y|U}(y|0) D_1(y) dy \\ &= - \int p_{Y|U}(y|0) \frac{\partial}{\partial u} \log p_{Y|U}(y|u) \Big|_{u=0} dy \\ &= - \int \frac{\partial}{\partial u} p_{Y|U}(y|u) \Big|_{u=0} dy \\ &= - \frac{\partial}{\partial u} \int p_{Y|U}(y|u) dy \Big|_{u=0} \\ &= - \frac{\partial}{\partial u} (1) = 0. \end{aligned} \quad (53)$$

Similarly, the second term in (52) can be simplified by evaluating the second-order derivative in the definition of  $D_2(y)$  to obtain

$$\begin{aligned} \mathbf{E}[D_2(y)|u=0] &= \int \frac{1}{p_{Y|U}(y|0)} \left( \frac{\partial}{\partial u} p_{Y|U}(y|u) \Big|_{u=0} \right)^2 dy \\ &\quad - \int \frac{\partial^2}{\partial u^2} p_{Y|U}(y|u) \Big|_{u=0} dy. \end{aligned} \quad (54)$$

Now,

$$\begin{aligned} &\int \frac{\partial^2}{\partial u^2} p_{Y|U}(y|u) \Big|_{u=0} dy \\ &= \frac{\partial^2}{\partial u^2} \int p_{Y|U}(y|u) \Big|_{u=0} dy = \frac{\partial^2}{\partial u^2} (1) = 0. \end{aligned} \quad (55)$$